SmartGate: A Multi-Factor Login System for Campus Security

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Computer Science 2

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April 19, 2026

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CHAPTER I
INTRODUCTION

Problem Statement/Rationale

Inadequate campus security caused by inconsistent entry and exit logging.

Project Overview and Relevance

This project proposes improving the current security of PSHS-CBZRC by developing a campus access monitoring program and an identity verification system. This program is developed using Python and integrates database management and analysis as a way to track the entry and exit of students, faculty, custodians, and visitors within the campus at a given time to create a safe and reliable platform for scholars. Its functions consist of a login mechanism where users input their ID number and password to verify their identity, and upon successful authentication, the system will automatically record their entry and exit, ensuring proper documentation of who enters and exits the campus, while also keeping track of where scholars are, which could indirectly assist in easily identifying scholars who skip classes.

However, this is a fraction of the multiple layers of verification that this system contains, which includes password authentication, LRN backup verification, and, in special cases, the ability for a custodian or faculty member to verify whether the user is truly part of PSHS ot there to deliver items that the school needs. If this is not the case and it's a stranger trying to infiltrate the school, removal from campus is effectively immediate. Other than that, to keep data organized, the database separates user roles into three categories: the scholars, the faculty, and the custodians, with the scholars separated into two more roles, which are the interns and the externs. This is so that a structured database can be able to store user credentials and activity records, while maintaining organized datasets that staff can monitor to guarantee a truly safe learning environment.

PSHS has been known as the premier science high school in the Philippines, but it has never been known as the most secure school in the country. This doesn't matter, though, as Pisay's goal is to produce globally competitive scholars. Still, you cannot produce top-tier scholars when those who say the word "Extern" can instantly go on campus without checking any background information, which is why this project is relevant: this issue can endanger students, teachers, and anyone on campus caught in the crossfire. Therefore, replacing these inconsistent security protocols with a new and improved standardized digital system can reduce the risk of unauthorized entry via database management and even the CRUD method. Additionally, to make the school more secure, basic data analysis can assist staff in observing patterns during the peak hours of entry and exit on campus, such as during the Flag Ceremony and the Flag Retreat, which can give employees enough information to plan out security ahead of time and manage the campus overall.

In the end, the researchers' overall goal is to make PSHS-CBZRC a safer environment for its students, while highlighting the importance of data systems in real-world scenarios by using them to solve problems affecting the campus, shining light as to how relevant Computer Science really is.

Objectives

This project focuses on improving campus security by implementing a system that tracks who enters and exits the school. The team ensures that the students' and staff data are properly recorded to maintain order and safety within the campus. The project allows users/admins to respond to possible emergencies much more easily, especially by keeping accurate records.

Another feature of this project is password-based authentication. The team had decided to go for this system to ensure that no unverified or unregistered people could enter the building to avoid possible threats.

Additionally, the project covers password recovery. The team accounted for the instance where users may forget their passwords, which could restrict access even if not necessary. To address this, the project provides a recovery process that allows the registered users to still get access while also maintaining security.

The project also has comprehensive and secure logs of campus access activities. The project records entry and exit data along with the date and time. These logs support accountability and help identify any unusual or suspicious behavior within the campus.

Overall, a secure monitoring system, password recovery, and activity logging all in one project were made to create a safer campus environment. The team aims to protect the security and safety of students and staff while also having an efficient system.

CHAPTER II
FINAL PROGRAM

Methodology

The researchers who made SmartGate used the developmental research method to do the research to create it. They chose this way because it helps them design, make, test and improve a system that solves a problem. The main goal of the study was to make a computer system that can tell who the authorized users are and keep track of when they come and go from the campus.

The first thing the researchers did was figure out what the problem was. They saw that it takes a lot of time to monitor students and staff when they enter and leave the campus, and it can be messy and prone to mistakes. The old way of using logbooks can lead to records that are hard to read and delays when it is busy. So the researchers suggested a system that uses passwords to recognize users, which can make monitoring faster and more organized.

The next step was to plan everything out. The researchers talked about what they wanted the system to do, what features it should have, and how it should work. They wanted to keep records for students and staff, and they needed to store important information like names, roles, birthdays, passwords, and class numbers. They also wanted to have a feature for administrators to add, remove, and check user records.

Then the researchers designed the system. They used a programming language called Python to make the system. They used lists to store all the user data. This made it easy to store and get all the information. They also made functions for important tasks like checking passwords, recovering passwords, and keeping track of time. This made the system more organized and easier to improve.

After that, the researchers wrote the code for the system. Made it work. When someone uses the system, it asks if they are coming or going from the campus. Then it asks for their password. If the password is correct, the system shows the user's information to confirm who they are. If the password is not correct, the user can try to recover their password using the LRN option.

The researchers then tested the system times with different user inputs. They tried passwords to see if the system could identify users, and they tried incorrect passwords to see if the recovery system worked. They also checked to make sure the system was recording the date and time when users came and went. They tested the administrator controls to make sure they could add, remove, and search records correctly.

The next step was to evaluate the system and make any changes. The researchers found some errors and weaknesses during testing. They fixed them. They made some adjustments to make the system more accurate, easy to use, and efficient. They improved the prompts, fixed some inputs, and made the system easier for users to understand.

Finally, the Password Recognition System with Time Monitoring for Campus Access was. It worked well. The system provides a more organized and more reliable way to record the movement of students and staff on the campus. The researchers were able to make a practical solution for monitoring the campus by using this special way of doing research. The Password Recognition System can really help the campus by keeping track of who's coming and going, and it can make things easier for everyone.

Flowchart and Program

Program in Python:

```
# AA of da Nerd Clusters :D
# This program is a password recognition system that allows
# students and staff to log in or out of the campus.
# It also includes admin controls and password recovery via LRN.

import datetime
from zoneinfo import ZoneInfo

# ========================
# DATA STORAGE (USING DICTIONARIES)
# ========================

# Lists for the Staff
slist = [
    {"name": "Guard, Frontman", "role": "Teacher/intern", "bday":
"27011984", "password": "twunc"}
]

# Lists for the students
students = [
    {"name": "A", "role": "student/intern", "bday": "06102012",
"password": "password",
     "grade": "8", "section": "Waling", "sectno": "204", "classno": "01"},
    {"name": "B", "role": "student/extern", "bday": "23042012",
"password": "0123456",
     "grade": "8", "section": "Waling", "sectno": "204", "classno": "67"}
]

# Lists that store login and logout times
time_entered = {}
time_exited = {}

# Temporary variables used to store current user info
tname = "0"
trole = "0"
tbday = "0"
tgrade = "0"
```

```
tsect = "0"
idno = -1
secno = "0"
clsno = "0"
yn2 = "0"

# ========================
# TIME TRACKING FUNCTION
# ========================
def time_check(action, user_id):
    current_time =
datetime.datetime.now(ZoneInfo("Asia/Manila")).strftime("%Y-%m-%d %I:%M:%S
%p")
    if action == "enter":
        time_entered[user_id] = current_time
    elif action == "exit":
        time_exited[user_id] = current_time

# ========================
# PASSWORD AUTHENTICATION
# ========================
def pswd_enter():

    global yn2, tname, trole, tbday, tgrade, tsect, idno

    # Reset temporary variables
    tname = "0"
    trole = "0"
    tbday = "0"
    tgrade = "0"
    tsect = "0"
    idno = -1

    pawd = input("Please enter your password: ")

    # ========================
    # STUDENT PASSWORD CHECK
    # ========================
    for i, student in enumerate(students):
        if pawd == student["password"]:
            idno = i
            tname = student["name"]
            trole = student["role"]
            tbday = student["bday"]
            tgrade = student["grade"]
            tsect = student["section"]
            return "student"
```

```
    # ========================
    # STAFF PASSWORD CHECK
    # ========================
    for i, staff in enumerate(slist):
        if pawd == staff["password"]:
            idno = i
            tname = staff["name"]
            trole = staff["role"]

            # Staff/Admin menu
            yn1 = int(input("0 - Add a person \n1 - Remove a person \n2 -
Manually enter \n3 - Check log in/log out time \nChoice: "))
            if yn1 == 0:
                sos = int(input("0 - Student \n1 - Staff \nChoice: "))
                nname = input("Enter the student/staff's name: ").title()
                npawd = input("Enter the student/staff's password: ")
                nbday = input("Enter the student/staff's birthday: ")
                nrole = input("Enter the student/staff's role:
").capitalize()
                if sos == 0:
                    ngrade = input("Enter grade: ")
                    nsect = input("Enter section name: ")
                    nsectno = input("Enter section number: ")
                    nclassno = input("Enter class number: ")
                    while nrole not in ["Student/extern",
"Student/intern"]:
                        nrole = input("Enter your student's role again:
").capitalize()
                    students.append({
                        "name": nname, "role": nrole, "bday": nbday,
"password": npawd,
                        "grade": ngrade, "section": nsect, "sectno":
nsectno, "classno": nclassno
                    })
                else:
                    while nrole not in ["Teacher/extern",
"Teacher/intern", "Custodian"]:
                        nrole = input("Enter your staff's role again:
").capitalize()
                    slist.append({"name": nname, "role": nrole, "bday":
nbday, "password": npawd})
                return "staff"
            elif yn1 == 1:
                ridno = int(input("Enter the ID number of the
student/staff you want to remove: "))
                sos2 = int(input("0 - Student \n1 - Staff\nChoice: "))
```

```
                if sos2 == 0 and 0 <= ridno < len(students):
                    removed = students.pop(ridno)
                    print(f"Removed student: {removed['name']}")
                elif sos2 == 1 and 0 <= ridno < len(slist):
                    removed = slist.pop(ridno)
                    print(f"Removed staff: {removed['name']}")
                return "staff"
            elif yn1 == 2:
                tname = input("Enter the name of the student: ").title()
                trole = input("Enter the role of the student:
").capitalize()
                tbday = input("Enter the birthday of the student: ")
                tidno = int(input("Enter the ID no. of the student: "))
                while True:
                    if 0 <= tidno < len(students) and tname ==
students[tidno]["name"] and \
                       trole == students[tidno]["role"] and tbday ==
students[tidno]["bday"]:
                        idno = tidno
                        return "staff"
                    else:
                        print("Incorrect details. Try again.")
                        tname = input("Enter the name of the student:
").title()
                        trole = input("Enter the role of the student:
").capitalize()
                        tbday = input("Enter the birthday of the student:
")
                        tidno = int(input("Enter the ID no. of the
student: "))
            elif yn1 == 3:
                while True:
                    search = input("Enter the student's/staff's name to
see their log in/log out time: ").title()
                    found = False
                    for i, student in enumerate(students):
                        if student["name"] == search:
                            print(f"Name: {student['name']} \nRole:
{student['role']} \nBirthday: {student['bday']}"
                                  f"\nGrade and Section:
{student['grade']} - {student['section']}"
                                  f"\nLogin time: {time_entered.get(i,'No
login record')}")
                            found = True
                            break
                    for i, staff in enumerate(slist):
                        if staff["name"] == search:
```

```
                            print(f"Name: {staff['name']} \nRole:
{staff['role']} \nBirthday: {staff['bday']}"
                                  f"\nLogin time: {time_entered.get(i,'No
login record')}")
                            found = True
                            break
                    if not found:
                        print("Name not found.")
                    done = input("Do you NOT want to search another
person? (Y/N): ").upper()
                    if done == "Y":
                        break
                return "staff"

    # ========================
    # INVALID PASSWORD
    # ========================
    print("The password you placed does not match any passwords in our
database.")
    yn2 = input("Do you want to use your LRN? (Y/N): ").upper()
    return None

# ========================
# MAIN PROGRAM LOOP
# ========================
print("=====PASSWORD RECOGNITION SYSTEM=====")

while True:
    # Ask user if entering or exiting
    useract = input("Type 'enter' if you are entering or type 'exit' if
you are exiting the campus: ").lower()

    # Ask for password
    result = pswd_enter()

    # If user wants LRN recovery
    if yn2 == "Y":
        while True:
            lrn_input = input("Enter your LRN: ")
            clsno = lrn_input[0:2]
            secno = lrn_input[2:5]
            tbday = lrn_input[5:13]

            # Search for student
            found = False
            for i, student in enumerate(students):
```

```
                if student["classno"] == clsno and student["sectno"] ==
secno and student["bday"] == tbday:
                    tname = student["name"]
                    trole = student["role"]
                    tgrade = student["grade"]
                    tsect = student["section"]
                    idno = i
                    found = True
                    print(f"Password recovered successfully for {tname}.")
                    break
            if found:
                break
            else:
                print("Invalid LRN. Try again.")

    # Authentication failed
    if idno == -1:
        print("Authentication failed. Please try again.\n")
        continue

    # Confirm identity
    yn3 = input(
        f"Is this you?: \n"
        f"Name: {tname} \n"
        f"Role: {trole} \n"
        f"Birthday: {tbday} \n"
        f"Grade and Section: {tgrade} - {tsect} \n"
        f"(Y/N): "
    ).upper()
    if yn3 == "N":
        print("Let's try again.\n")
        continue

    # Record time if correct
    if useract in ["enter", "exit"]:
        time_check(useract, idno)
        print("Time recorded successfully!\n")
    else:
        print("Invalid action. Please type 'enter' or 'exit'.\n")
        continue

    # Ask if user wants to restart
    yn4 = input("Do you want to start again? (Y/N): ").upper()
    if yn4 == "N":
        break

# ========================
```

```
# ATTENDANCE / LOGIN-LOGOUT TABLE
# ========================
print("\n===== CAMPUS LOGIN/LOGOUT RECORDS =====")
print(f"{'ID':<4}{'Name':<20}{'Role':<20}{'Login Time':<25}{'Logout
Time':<25}")
print("-"*94)

# Show students
for i, student in enumerate(students):
    login = time_entered.get(i, "No record")
    logout = time_exited.get(i, "No record")

print(f"{i:<4}{student['name']:<20}{student['role']:<20}{login:<25}{logout
:<25}")

# Show staff
offset = len(students)  # Staff IDs after students
for i, staff in enumerate(slist):
    login = time_entered.get(i + offset, "No record")
    logout = time_exited.get(i + offset, "No record")
    print(f"{i +
offset:<4}{staff['name']:<20}{staff['role']:<20}{login:<25}{logout:<25}")

print("\nSystem closed.")
```

Sample Output

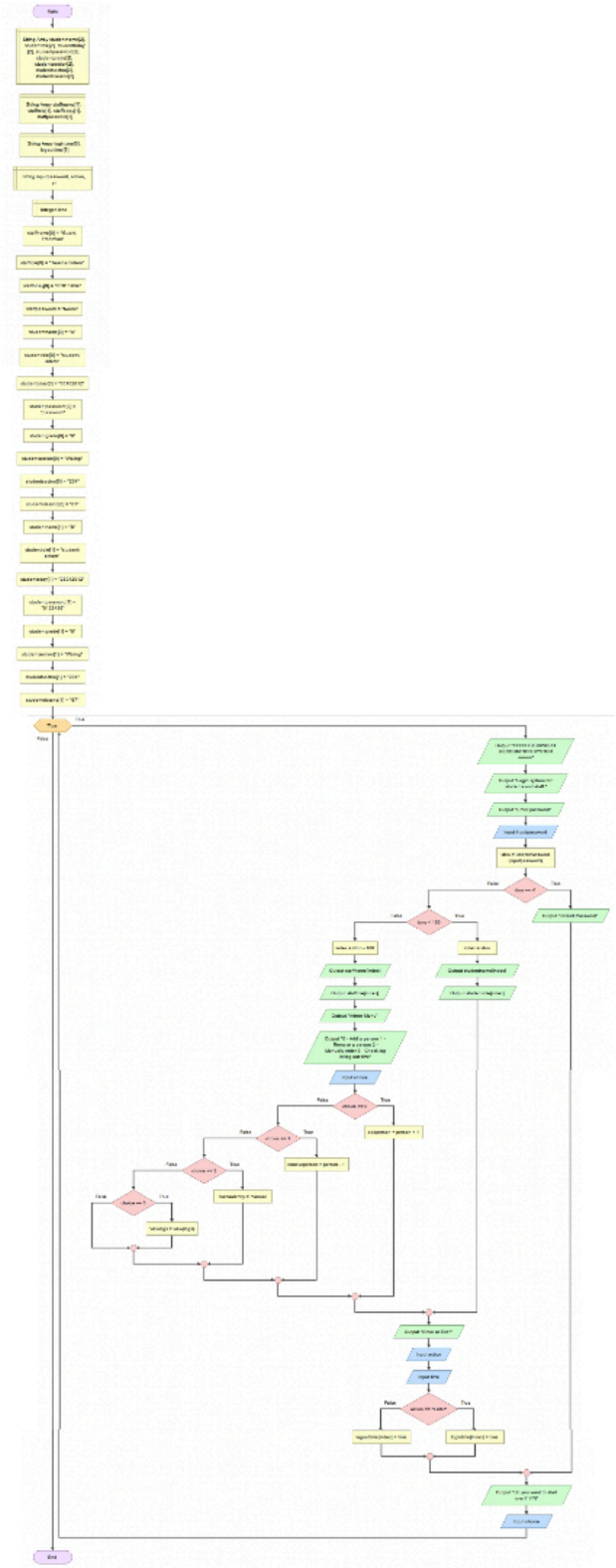
```
=====PASSWORD RECOGNITION SYSTEM=====
Type 'enter' if you are entering or type 'exit' if you are exiting the
    campus: enter
Please enter your password: password
Is this you?:
Name: A
Role: student/intern
Birthday: 06102012
Grade and Section: 8 - Waling
(Y/N): Y
Time recorded successfully!

Do you want to start again? (Y/N):
```

```
=====PASSWORD RECOGNITION SYSTEM=====
Type 'enter' if you are entering or type 'exit' if you are exiting the
    campus: enter
Please enter your password: enter
The password you placed does not match any passwords in our database.
Do you want to use your LRN? (Y/N):
```

```
=====PASSWORD RECOGNITION SYSTEM=====
Type 'enter' if you are entering or type 'exit' if you are exiting the
    campus: enter
Please enter your password: twunc
0 - Add a person
1 - Remove a person
2 - Manually enter
3 - Check log in/log out time
Choice: 3
Enter the student's/staff's name to see their log in/log out time: A
Name: A
Role: student/intern
Birthday: 06102012
Grade and Section: 8 - Waling
Login time: No login record
Do you NOT want to search another person? (Y/N): N
```

Flowchart



```
<?xml version="1.0"?>
<flowgorithm fileversion="3.0">
    <attributes>
        <attribute name="name" value=""/>
        <attribute name="authors" value="yuri"/>
        <attribute name="about" value=""/>
        <attribute name="saved" value="2026-04-19 04:52:05 PM"/>
        <attribute name="created"
value="eXVyaTtTLUxBUFRPUDsyMDI2LTA0LTE5OzAzOjQ4OjQ4IFBNOzIzNDQ="/>
        <attribute name="edited"
value="eXVyaTtTLUxBUFRPUDsyMDI2LTA0LTE5OzA0OjUyOjA1IFBNOzQ7MjQ0NA=="/>
    </attributes>
    <function name="Main" type="None" variable="">
        <parameters/>
        <body>
            <declare name="studentname, studentrole, studentbday,
studentpassword, studentgrade, studentsection, studentsectno,
studentclassno" type="String" array="True" size="2"/>
            <declare name="staffname, staffrole, staffbday, staffpassword"
type="String" array="True" size="1"/>
            <declare name="logintime, logouttime" type="String"
array="True" size="3"/>
            <declare name="inputpassword, action, yn" type="String"
array="False" size=""/>
            <declare name="idno" type="Integer" array="False" size=""/>
            <assign variable="staffname[0]" expression="&quot;Guard,
Frontman&quot;"/>
            <assign variable="staffrole[0]"
expression="&quot;Teacher/intern&quot;"/>
            <assign variable="staffbday[0]"
expression="&quot;27011984&quot;"/>
            <assign variable="staffpassword"
expression="&quot;twunc&quot;"/>
            <assign variable="studentname[0]" expression="&quot;A&quot;"/>
            <assign variable="studentrole[0]"
expression="&quot;student/intern&quot;"/>
            <assign variable="studentbday[0]"
expression="&quot;06102012&quot;"/>
            <assign variable="studentpassword[0]"
expression="&quot;password&quot;"/>
            <assign variable="studentgrade[0]"
expression="&quot;8&quot;"/>
            <assign variable="studentsection[0]"
expression="&quot;Waling&quot;"/>
            <assign variable="studentsectno[0]"
expression="&quot;204&quot;"/>
```

```
            <assign variable="studentclassno[0]"
expression="&quot;01&quot;"/>
            <assign variable="studentname[1]" expression="&quot;B&quot;"/>
            <assign variable="studentrole[1]"
expression="&quot;student/extern&quot;"/>
            <assign variable="studentbday[1]"
expression="&quot;23042012&quot;"/>
            <assign variable="studentpassword[1]"
expression="&quot;0123456&quot;"/>
            <assign variable="studentgrade[1]"
expression="&quot;8&quot;"/>
            <assign variable="studentsection[1]"
expression="&quot;Waling&quot;"/>
            <assign variable="studentsectno[1]"
expression="&quot;204&quot;"/>
            <assign variable="studentclassno[1]"
expression="&quot;67&quot;"/>
            <while expression="True">
                <output expression="&quot;===== PASSWORD RECOGNITION
SYSTEM =====&quot;" newline="True"/>
                <output expression="&quot;Login system for students and
staff.&quot;" newline="True"/>
                <output expression="&quot;Enter password&quot;"
newline="True"/>
                <input variable="inputpassword"/>
                <assign variable="idno"
expression="CheckPassword(inputpassword)"/>
                <if expression="idno == -1">
                    <then>
                        <output expression="&quot;Invalid Password&quot;"
newline="True"/>
                    </then>
                    <else>
                        <if expression="idno &lt; 100">
                            <then>
                                <assign variable="index"
expression="idno"/>
                                <output expression="studentname[index]"
newline="True"/>
                                <output expression="studentrole[index]"
newline="True"/>
                            </then>
                            <else>
                                <assign variable="index" expression="idno
- 100"/>
                                <output expression="staffname[index]"
newline="True"/>
```

```
                                <output expression="staffrole[index]"
newline="True"/>
                                <output expression="&quot;Admin
Menu&quot;" newline="True"/>
                                <output expression="&quot;0 - Add a person
1 - Remove a person 2 - Manually enter 3 - Check log in/log out
time&quot;" newline="True"/>
                                <input variable="choice"/>
                                <if expression="choice == 0">
                                    <then>
                                        <assign variable="addperson"
expression="person + 1"/>
                                    </then>
                                    <else>
                                        <if expression="choice == 1">
                                            <then>
                                                <assign
variable="removeperson" expression="person - 1"/>
                                            </then>
                                            <else>
                                                <if expression="choice ==
2">
                                                    <then>
                                                        <assign
variable="manualentry" expression="manual"/>
                                                    </then>
                                                    <else>
                                                        <if
expression="choice == 3">
                                                            <then>
                                                                <assign
variable="viewlogs" expression="view(logs)"/>
                                                            </then>
                                                            <else/>
                                                        </if>
                                                    </else>
                                                </if>
                                            </else>
                                        </if>
                                    </else>
                                </if>
                            </else>
                        </if>
                        <output expression="&quot;Enter or Exit?&quot;"
newline="True"/>
                        <input variable="action"/>
                        <input variable="time"/>
```

```
                        <if expression="action == &quot;Enter&quot;">
                            <then>
                                <assign variable="logintime[index]"
expression="time"/>
                            </then>
                            <else>
                                <assign variable="logouttime[index]"
expression="time"/>
                            </else>
                        </if>
                    </else>
                </if>
                <output expression="&quot;Do you want to start over?
Y/N&quot;" newline="True"/>
                <input variable="choice"/>
            </while>
        </body>
    </function>
    <function name="CheckPassword" type="Integer" variable="idno">
        <parameters>
            <parameter name="password" type="String" array="False"/>
        </parameters>
        <body>
            <declare name="i, idno" type="Integer" array="False" size=""/>
            <assign variable="idno" expression="-1"/>
            <for variable="i" start="0" end="1" direction="inc" step="1">
                <if expression="password == studentpassword[i]">
                    <then>
                        <assign variable="idno" expression="1"/>
                    </then>
                    <else/>
                </if>
            </for>
            <for variable="i" start="0" end="0" direction="inc" step="1">
                <if expression="password == staffpassword[i]">
                    <then>
                        <assign variable="idno" expression="1+100"/>
                    </then>
                    <else/>
                </if>
            </for>
        </body>
    </function>
</flowgorithm>
```

CHAPTER III
CONCLUSION

Team Roles

MEMBERS	Q1	Q2	Q3	Q4
Aquino, Yuri Oliva	Did the scope and limitations	Added most of the essential requirements	Assisted in fixing the code	Did the flowchart and readme.md
Balondo, Mikah Gabrielle	Did the problem identification	Did the problem decomposition	Did the readme	Did the problem statement and assisted with the objectives
Batecan, Maria Cielo	Assisted in scope and limitations	Assisted in the problem decomposition	Did the readme	Did the objectives
Fajilan, Joaquin Ryan	Did the suggested solutions	N/A	Did the whole program coding	Did the methodology and code
Macasaet, Athena Ingrid	Did the problem statement	Did the whole flowchart	Did the readme	Did the project overview and relevance

References